

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			bromothymol blue and naphtholphthalein (1) change colour completely within vertical range of curve (1)	1	1		2		2
	(b)			moles NaOH = $20.0 \times \frac{0.250}{1000} = 5.00 \times 10^{-3}$ (1) $[HA] = \frac{5.00 \times 10^{-3}}{25.0 \times 10^{-3}} = 0.200 \text{ mol dm}^{-3}$ (1)		2		2	1	2
	(c)			$pK_a = \text{pH at half neutralisation} = 3.85 (\pm 0.1)$ (1) $K_a = 1.41 \times 10^{-4} \text{ mol dm}^{-3}$ (1) ECF possible for any value in the range 1.12×10^{-4} to 1.78×10^{-4} $[H^+]^2 = K_a \times [HA] = 2.83 \times 10^{-5}$ (1) ECF possible for any value in the range 2.24×10^{-5} to 3.56×10^{-5} $\text{pH} = 2.27$ (1) ECF possible for any value in the range 2.22 to 2.32		1 1	1 1	4	3	
	(d)			award (1) for any of following storing/using enzymes (at constant pH) fermentation dyeing accept any other sensible use	1			1		

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	(e)			award (1) for rearranged equation $T = \frac{\Delta H - \Delta G}{\Delta S}$ $T = 307\text{ K} \quad (1)$ process is exothermic / enthalpy change is negative so when temperature increases equilibrium shifts to the left (1) accept converse answer if temperature calculated is below 298K		1	2	3	2	
				Question 10 total	2	6	4	12	6	4